

SUMMARY

This video explains the core principles of Deterministic Finite Automata (DFA) construction, with emphasis on minimal state design for pattern-based languages. It distinguishes among “Starts with,” “Ends with,” and “Substring” constructions and formalizes architectural rules governing state complexity.

Key conclusions:

Minimal DFA uses the smallest possible number of states.

Substring patterns generally follow the $n + 1$ rule.

Dead states strongly indicate “Starts with” logic.

Final-state self-loops distinguish substring DFAs from suffix DFAs.

I. FUNDAMENTAL PRINCIPLES OF MINIMAL DFA CONSTRUCTION

I.1 Definition of Minimal DFA

A DFA is minimal when:

It recognizes the language correctly.

It contains no unreachable states.

It contains no equivalent states.

It uses the minimum number of states possible.

I.2 Completeness and Determinism

In a DFA:

Every state must define exactly one transition for each symbol in Σ .

For binary alphabet $\Sigma = \{0, 1\}$:

Each state must have one 0-transition and one 1-transition.

Missing transitions imply nondeterminism, resulting in an NFA rather than a DFA.

I.3 The $n + 1$ Rule for Substrings

If language L is defined as “contains substring S ” where $|S| = n$:

Minimal States = $n + 1$

Examples:

Substring Length	Minimal States
2 (e.g., 00)	3
3 (e.g., 010)	4
4 (e.g., 1011)	5

II. COMPARATIVE ANALYSIS OF DFA CATEGORIES

Different pattern requirements produce structurally distinct DFAs.

Feature	Starts With	Ends With	Substring
Dead State	Compulsory	Not typical	Not typical
Final State Self-Loop	Present	Never present	Always present

Core Logic	Validates first n symbols	Tracks last n symbols	Locks once pattern found
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III. SUBSTRING CONSTRUCTION LOGIC

Substring DFAs move progressively through states that represent partial matches of the pattern.

III.1 Substring "0"

States: 2

State Structure:

Q_0 : Initial

Q_1 : Final

Transitions:

From Q_0 :

$0 \rightarrow Q_1$

$1 \rightarrow Q_0$

From Q_1 :

$0 \rightarrow Q_1$

$1 \rightarrow Q_1$

Once "0" is found, the DFA remains in final state permanently.

III.2 Substring "10"

States: 3

Structure:

Q_0 : No match yet

Q_1 : Seen '1'

Q_2 : Final (seen "10")

Transitions ensure that:

Incorrect continuation resets appropriately.

Once "10" is found, Q_2 self-loops on both inputs.

III.3 Substring "11"

States: 3

Key insight:

If at state representing single '1' and input '0' occurs, reset to Q_0 because consecutive pattern is broken.

IV. COMPLEX START AND END CONDITIONS

IV.1 Starts and Ends with Same Symbol

Requirement:

String must begin and end with same symbol.

State Count: 5

Logic:

Branch at initial state depending on first symbol.

Track final symbol dynamically.

Single-character strings must be accepted.

IV.2 Starts and Ends with Different Symbols

State Count: 5

If first symbol is 0 → must end in 1.
 If first symbol is 1 → must end in 0.
 Final states shift accordingly from same-symbol logic.

IV.3 Ends with “00” or “11”

Condition:

Last two symbols identical.

State Count: 5

Architecture:

Track previous symbol.

If next symbol matches previous, move to final state.

Continue updating tracking to preserve determinism.

V. EXAMINATION STRATEGY (GATE FOCUS)

V.1 The 5-Second Identification Rule

Architectural Feature	Likely Language Type
Final state has self-loop (0,1)	Substring DFA
No dead state, no final self-loop	Ends With DFA
Presence of dead state	Starts With DFA

V.2 Complement Construction

To build DFA for:

“Does not contain pattern X”

Procedure:

Construct DFA for “Contains X.”

Swap final and non-final states.

V.3 Pattern Recognition Cues

Phrase in Question	Interpretation
“Containing the pattern”	Substring
“Ends with”	Suffix logic
“Starts with”	Prefix validation

VI. HISTORICAL EXAM REFERENCES

Year	Topic	Insight
2009	Identify DFA language	Lack of final self-loop indicated suffix DFA
2021	Ends with “011”	Used $n + 1$ rule → 4 states
2025	Even number of b’s	Eliminated due to ϵ not accepted

1998	Last two symbols same	Required 5 states
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VII. TECHNICAL TERMINOLOGY

Term	Definition
Dibit	Substring of length 2 (e.g., 00, 11)
Tribit	Substring of length 3 (e.g., 000, 111)
Dead State	Non-final state from which no final state is reachable
Minimal DFA	DFA with minimum possible number of states recognizing the language